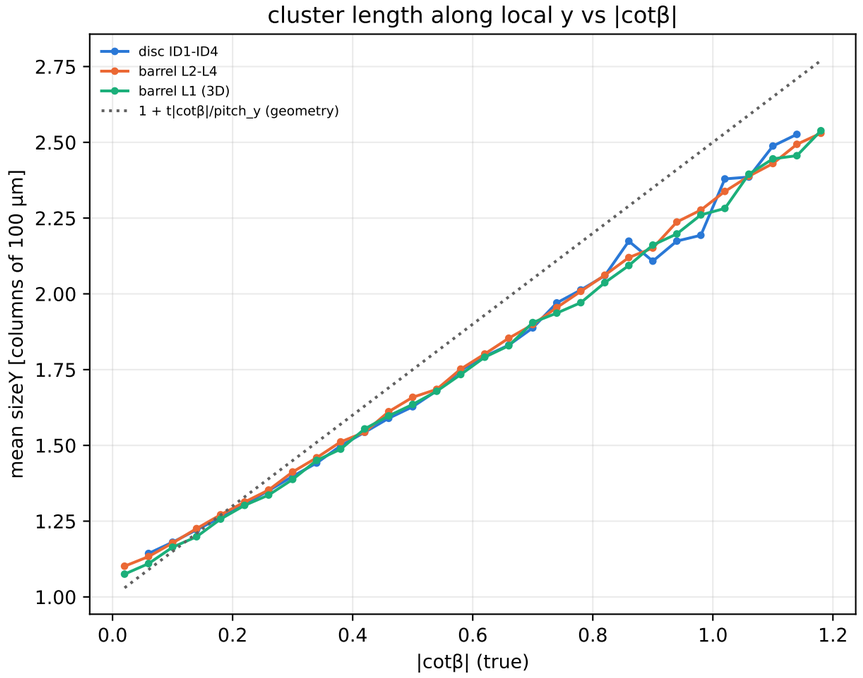
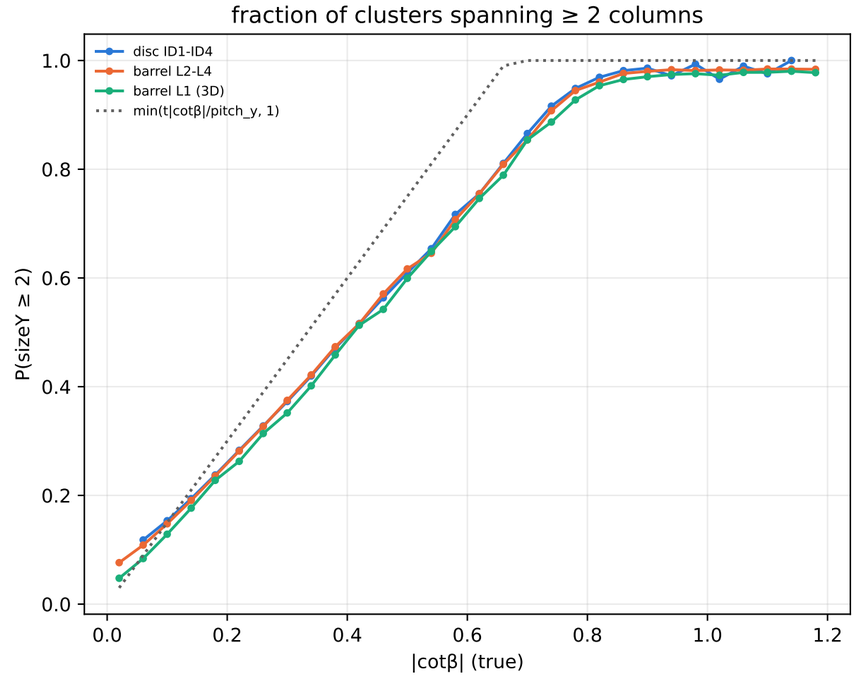
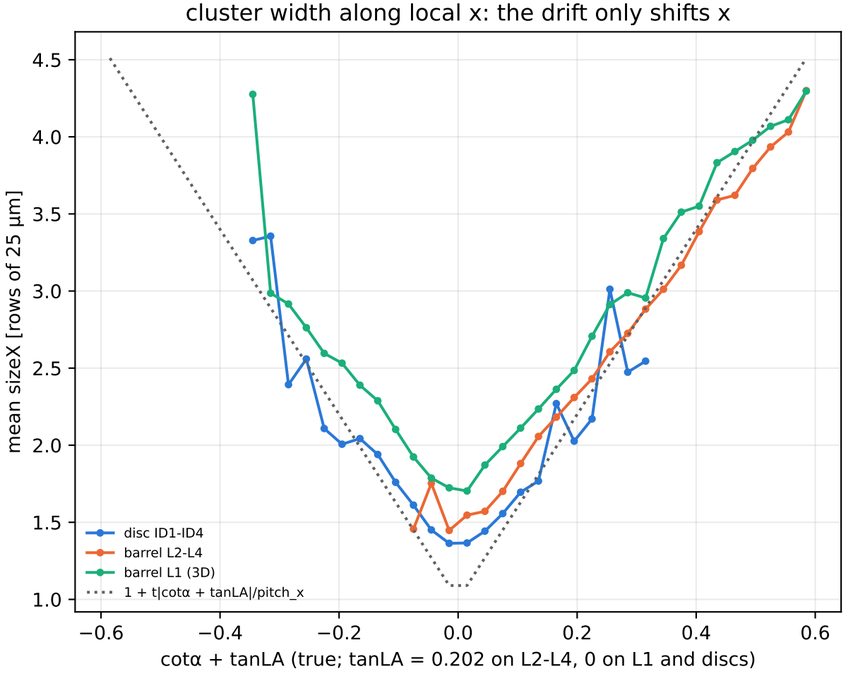
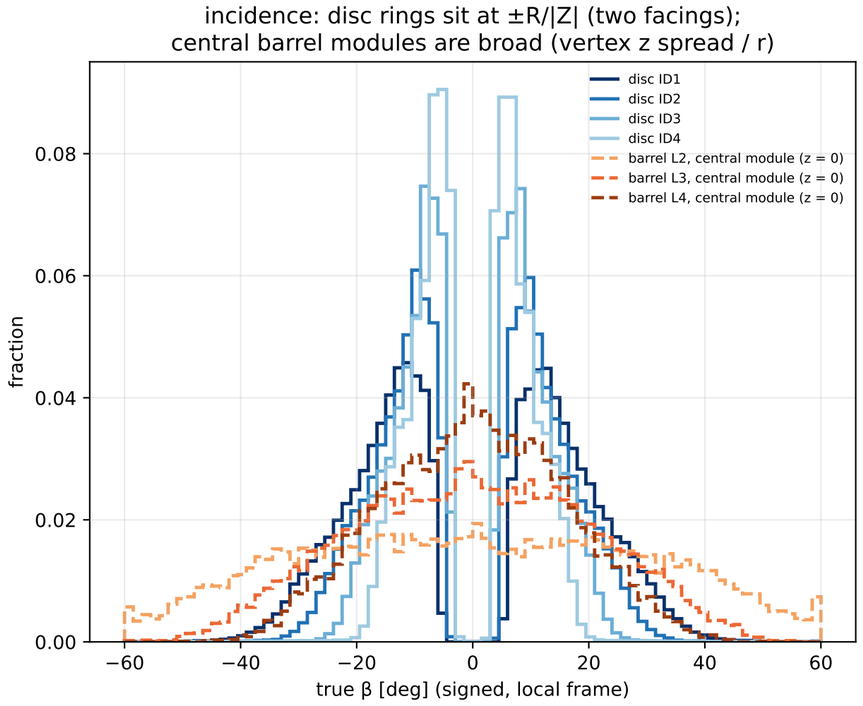
Page 1. Frames and cluster size vs incidence (consistent-linked, pT ≥ 1 GeV, 200 events; truth + clusters only)

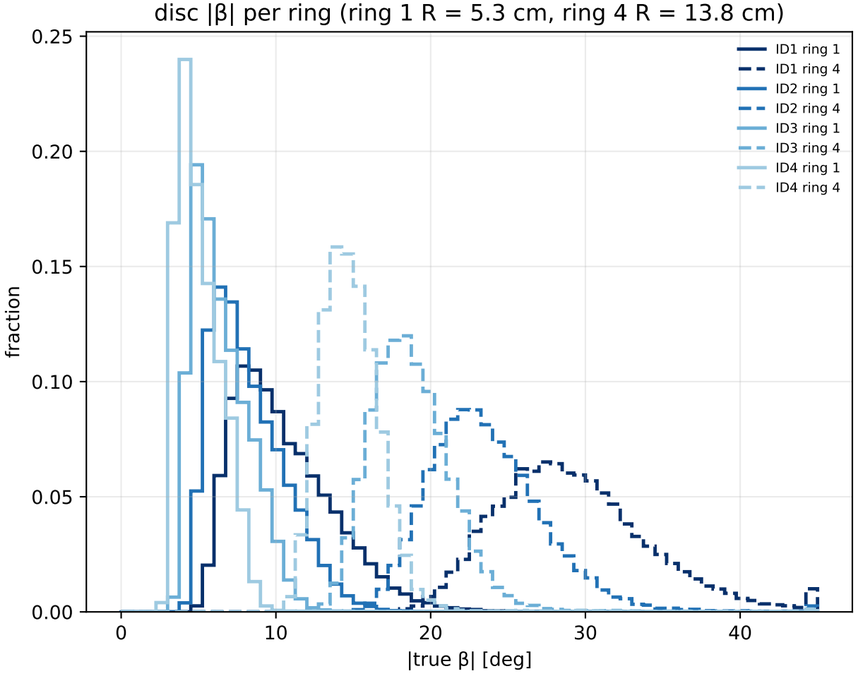
```
LOCAL FRAMES (all 4000 modules; module axes in global coordinates)
every module: pitch x 25 um, pitch y 100 um, thickness 150 um;
  charge collected at w = -t/2 (Phase-2 digitizer, dir_z = -1)
barrel: y = -z on every module; w = +r or -r (zOutward), half each;
  x = +-phi; drift along +x, tanLA 0.202 (L2-L4), 0 (L1 3D)
disc ID1-ID4: y = -r (radially inward) on every module;
  w = +z or -z (facing), half each per ring; x = +-phi; no drift
=> beta = the plane (100 um pitch, normal) = the r-z plane on BOTH
prompt cotBeta = -zOutward z/r (barrel), -facing R/|Z| (disc)
a flip = a rigid 180 deg rotation (x, w reversed, y kept):
  the image <-> local-angle relation is identical for both facings
```



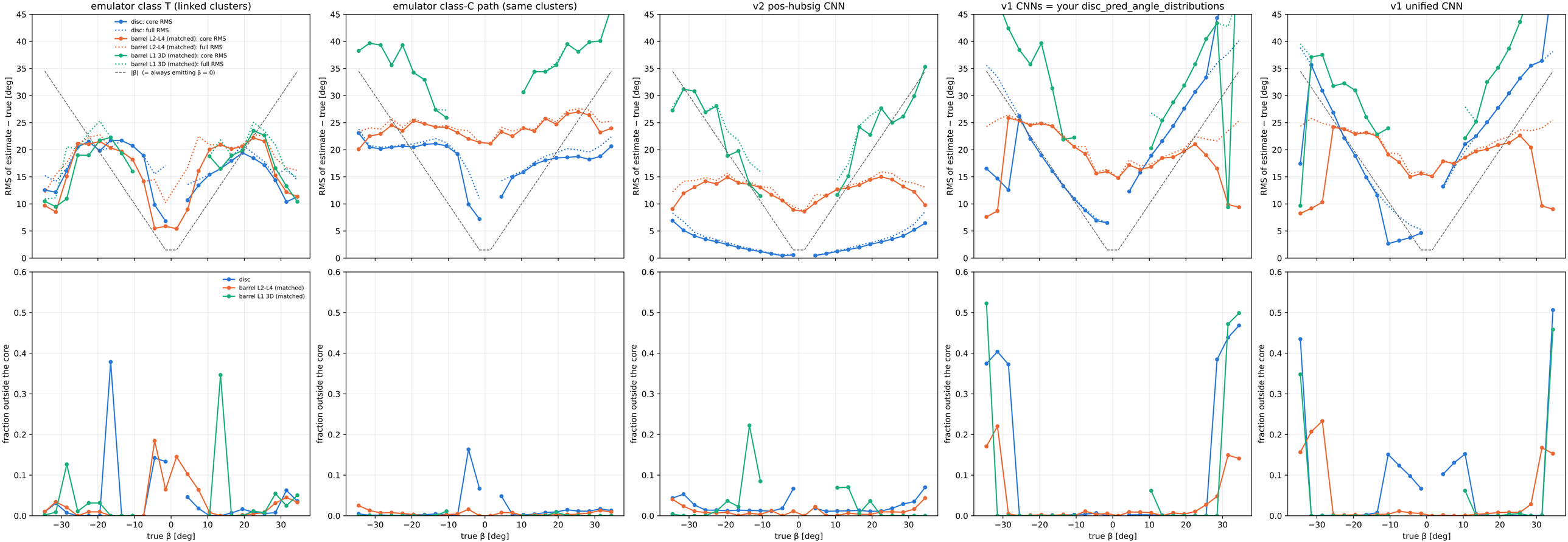




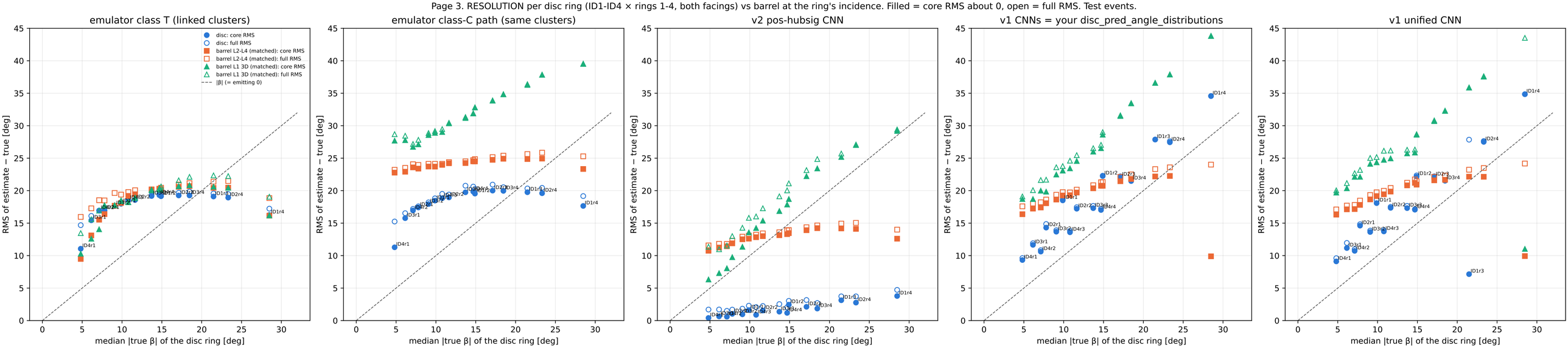




Page 2. RESOLUTION vs true β: discs ID1-ID4 (both facings) vs barrel clusters reweighted to the same true (β, α). Core RMS about ZERO (mode-seeded 3σ clip; bias counts), full RMS dotted;
bottom: fraction outside the core. 3° bins with effective N ≥ 30; consistent-linked, pT ≥ 1 GeV, the 50 test events. Where the residual is bimodal
(the class-T response near normal) the core latches onto one mode and the outside fraction jumps: the full RMS is the stable number there.



Page 3. RESOLUTION per disc ring (ID1-ID4 × rings 1-4, both facings) vs barrel at the ring's incidence. Filled = core RMS about 0, open = full RMS. Test events.



Resolution per disc ring vs matched barrel: core RMS about 0 [deg] (columns estimator:sample; test events, consistent-linked, pT ≥ 1)

disk	ring	|β|med	emuT:disc	emuT:L234	emuT:L1	emuC:disc	emuC:L234	emuC:L1	v2:disc	v2:L234	v2:L1	v1sep:disc	v1sep:L234	v1sep:L1	v1uni:disc	v1uni:L234	v1uni:L1
1	1	9.9	18.0	18.1	18.4	18.5	23.7	28.9	1.6	12.6	13.6	18.5	19.3	23.1	18.1	19.2	24.4
1	2	14.9	19.1	20.2	20.1	19.5	24.5	32.8	2.4	13.5	18.7	22.2	20.7	28.6	22.2	20.9	28.6
1	3	21.5	19.1	20.4	20.6	19.8	24.9	36.3	3.1	14.2	25.2	27.9	22.2	36.6	7.1	22.2	35.9
1	4	28.5	16.3	16.2	16.2	17.6	23.3	39.5	3.8	12.6	29.2	34.6	9.9	43.8	34.9	9.9	11.0
2	1	7.8	17.0	16.4	17.7	17.4	23.4	27.1	1.0	11.9	9.8	14.3	18.1	19.8	14.6	17.8	22.1
2	2	11.6	18.6	19.4	18.9	19.0	24.2	30.4	1.5	13.0	15.4	17.2	19.7	24.6	17.4	19.9	25.0
2	3	17.1	19.3	20.6	20.7	20.0	24.7	33.9	2.1	13.9	22.2	22.1	21.5	31.5	22.1	21.6	30.7
2	4	23.3	18.9	20.5	20.4	19.6	24.9	37.8	2.8	14.1	27.0	27.4	22.3	37.9	27.5	22.1	37.6
3	1	6.1	15.4	13.1	12.6	15.8	22.9	27.8	0.6	11.2	7.3	11.7	17.2	18.7	11.2	17.1	20.4
3	2	9.1	17.7	17.7	17.7	17.9	23.7	28.6	1.0	12.5	11.4	13.7	18.6	22.5	13.6	18.7	24.1
3	3	13.7	19.2	20.1	20.0	19.7	24.3	31.2	1.4	13.1	16.9	17.3	20.4	26.0	17.3	20.8	25.7
3	4	18.5	19.3	20.7	20.8	20.0	24.9	34.8	1.8	14.2	23.4	21.5	21.8	33.4	21.5	21.7	32.3
4	1	4.8	11.1	9.5	10.3	11.3	22.8	27.7	0.4	10.7	6.3	9.3	16.4	18.7	9.1	16.3	19.7
4	2	7.1	16.9	15.5	14.1	17.0	23.6	26.7	0.6	11.4	8.0	10.6	17.4	20.0	10.7	17.2	22.1
4	3	10.8	18.5	19.2	18.2	18.8	24.0	29.0	0.9	12.8	14.2	13.6	19.3	23.4	13.7	19.4	24.7
4	4	14.7	19.3	20.3	20.2	19.9	24.5	31.9	1.2	13.3	17.9	17.0	20.8	26.5	17.1	21.2	25.9

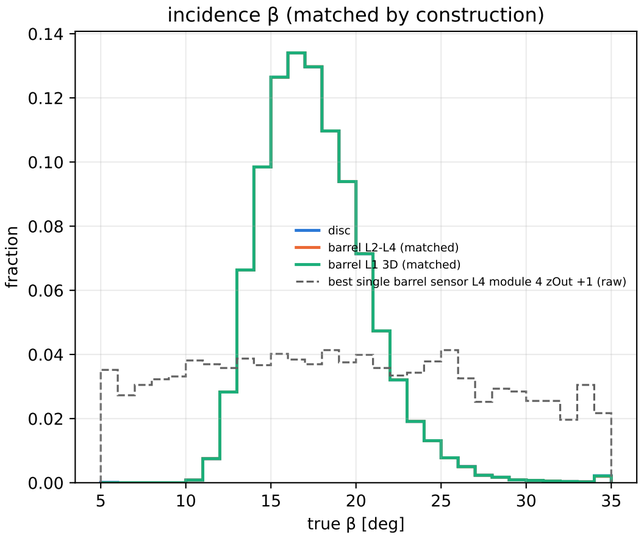
Resolution per disc ring vs matched barrel: full RMS [deg] (columns estimator:sample; test events, consistent-linked, pT ≥ 1)

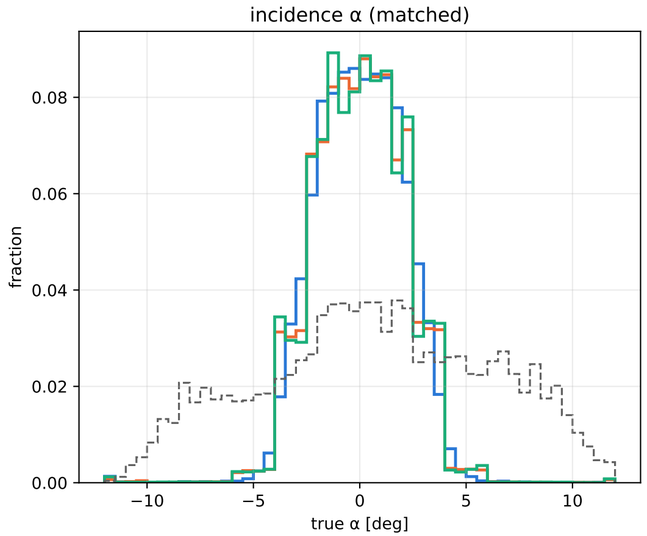
disk	ring	|β|med	emuT:disc	emuT:L234	emuT:L1	emuC:disc	emuC:L234	emuC:L1	v2:disc	v2:L234	v2:L1	v1sep:disc	v1sep:L234	v1sep:L1	v1uni:disc	v1uni:L234	v1uni:L1
1	1	9.9	18.0	19.4	18.5	18.8	24.1	29.2	2.3	13.1	15.8	18.6	19.8	23.8	18.6	19.7	25.2
1	2	14.9	19.4	20.5	20.6	20.3	24.9	32.8	3.0	13.9	21.1	22.3	21.3	29.0	22.3	21.5	28.7
1	3	21.5	19.6	21.4	22.3	20.3	25.6	36.4	3.7	14.9	25.7	27.9	23.3	36.6	27.9	23.2	35.9
1	4	28.5	17.2	18.9	19.1	19.2	25.3	39.6	4.7	14.0	29.4	34.6	24.0	43.8	34.9	24.2	43.5
2	1	7.8	17.0	18.5	17.8	17.6	24.0	27.8	1.7	12.3	13.0	14.9	18.6	21.7	14.8	18.5	23.2
2	2	11.6	18.6	20.1	18.9	19.4	24.4	30.4	2.2	13.4	17.3	17.5	20.2	25.5	17.5	20.4	26.2
2	3	17.1	19.7	20.9	21.6	20.9	25.2	33.9	3.2	14.4	23.2	22.2	22.1	31.6	22.2	22.1	30.8
2	4	23.3	19.5	21.5	22.2	20.4	25.8	37.9	3.7	15.0	27.1	27.6	23.6	37.9	27.6	23.5	37.6
3	1	6.1	16.1	17.3	15.6	16.5	23.5	28.5	1.7	11.8	11.0	11.9	18.0	20.1	12.0	17.7	21.2
3	2	9.1	17.7	19.6	17.8	18.2	24.1	28.9	1.8	12.9	14.3	13.9	19.4	23.6	13.9	19.3	25.0
3	3	13.7	19.3	20.2	20.2	20.8	24.5	31.3	2.5	13.6	19.1	17.7	20.8	26.5	17.7	21.2	26.3
3	4	18.5	19.9	21.2	22.1	20.5	25.4	34.9	2.7	14.6	24.9	21.6	22.5	33.5	21.6	22.3	32.3
4	1	4.8	14.7	16.0	13.5	15.3	23.2	28.7	1.7	11.6	11.4	9.6	17.6	19.1	9.6	17.1	20.1
4	2	7.1	16.9	18.5	16.9	17.2	24.1	27.2	1.5	11.8	11.5	10.8	18.2	21.6	11.0	17.8	22.7
4	3	10.8	18.6	19.8	18.2	19.5	24.1	29.5	2.1	13.2	16.0	13.8	19.7	24.6	13.9	20.0	26.1
4	4	14.7	19.5	20.3	20.3	20.6	24.7	31.9	2.0	13.9	20.0	17.2	21.3	27.1	17.1	21.7	26.3

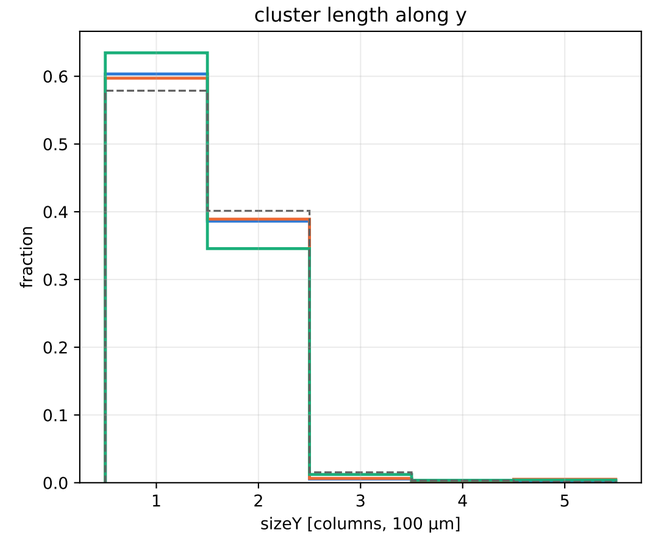
Resolution per disc ring vs matched barrel: fraction outside the core (columns estimator:sample; test events, consistent-linked, pT ≥ 1)

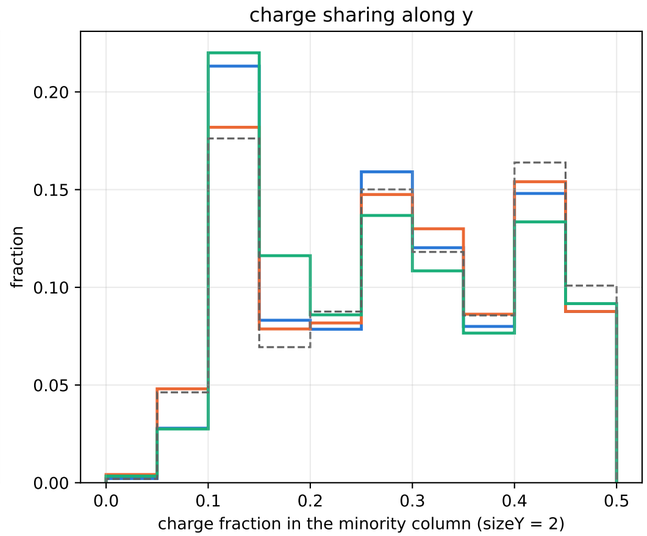
disk	ring	|β|med	emuT:disc	emuT:L234	emuT:L1	emuC:disc	emuC:L234	emuC:L1	v2:disc	v2:L234	v2:L1	v1sep:disc	v1sep:L234	v1sep:L1	v1uni:disc	v1uni:L234	v1uni:L1
1	1	9.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	2	14.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	3	21.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0
1	4	28.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.2	0.5
2	1	7.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	2	11.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	3	17.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	4	23.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	1	6.1	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.1	0.0	0.0
3	2	9.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	3	13.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	4	18.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	1	4.8	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0
4	2	7.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	3	10.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	4	14.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

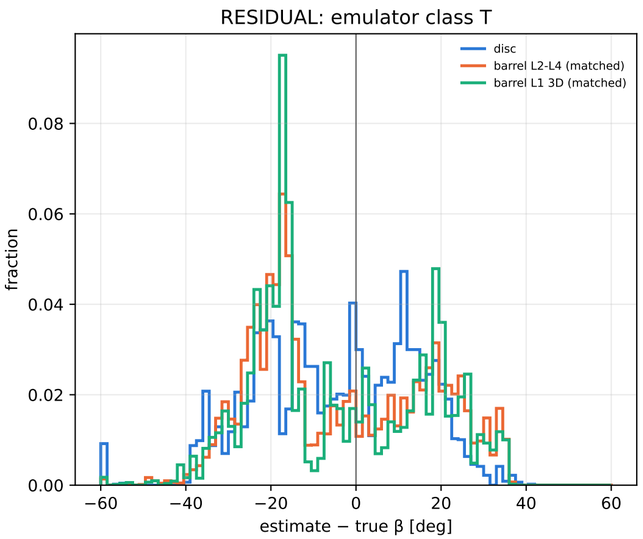
Page 4. Representative disc sensor ID2 ring 3 (R = 10.5 cm, |Z| = 32.9 cm), facing − (w toward the IP ⇒ β > 0) vs barrel sensors at MATCHED incidence (reweighted in true β × α). consistent-linked, pT ≥ 1 GeV

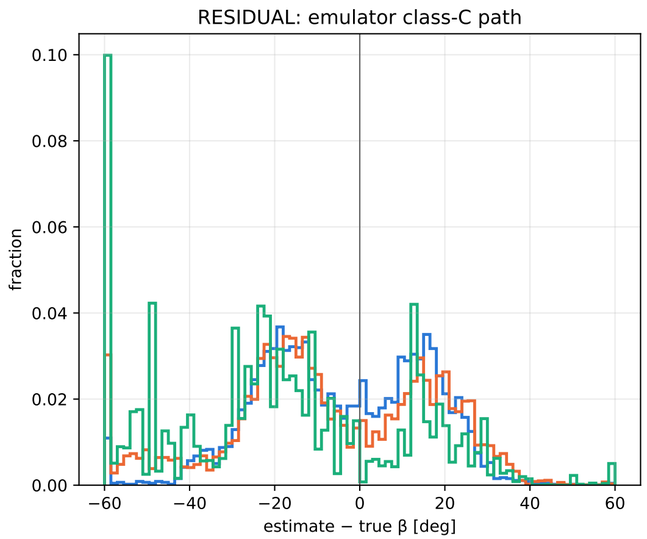


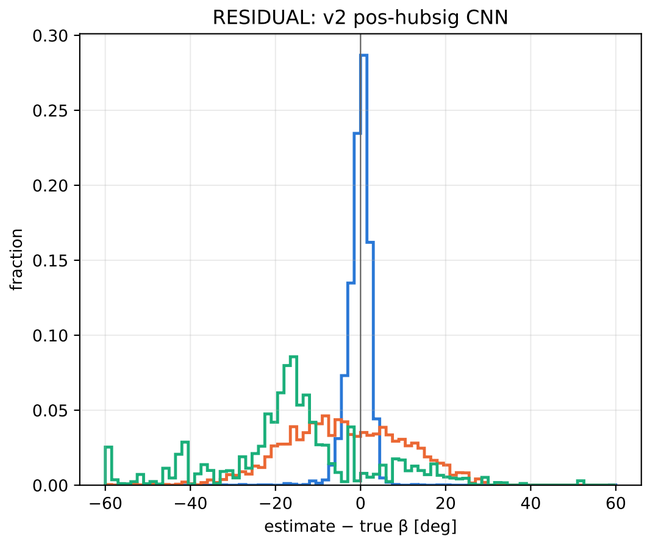


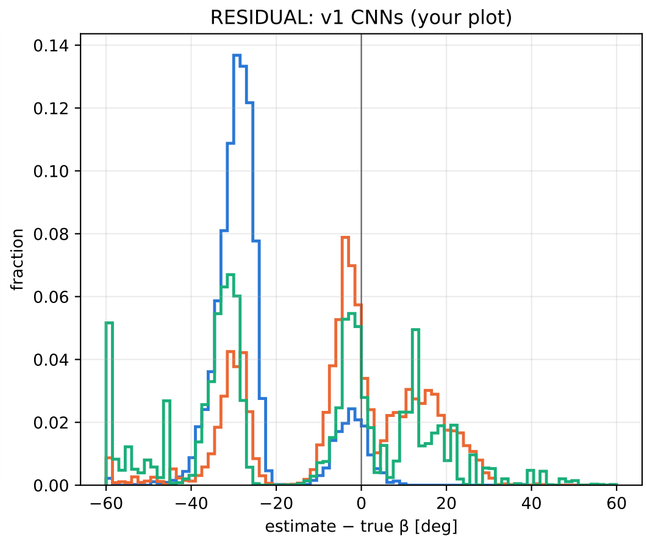


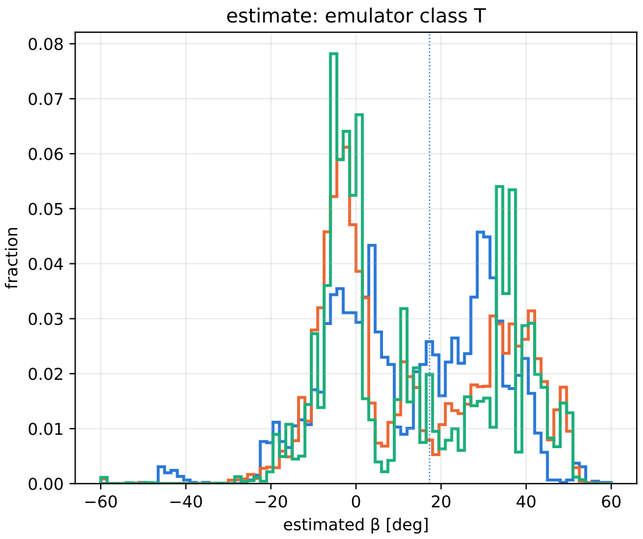


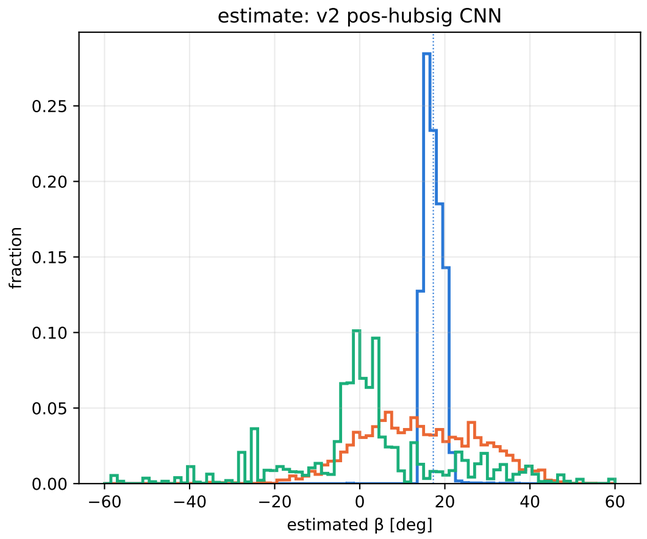


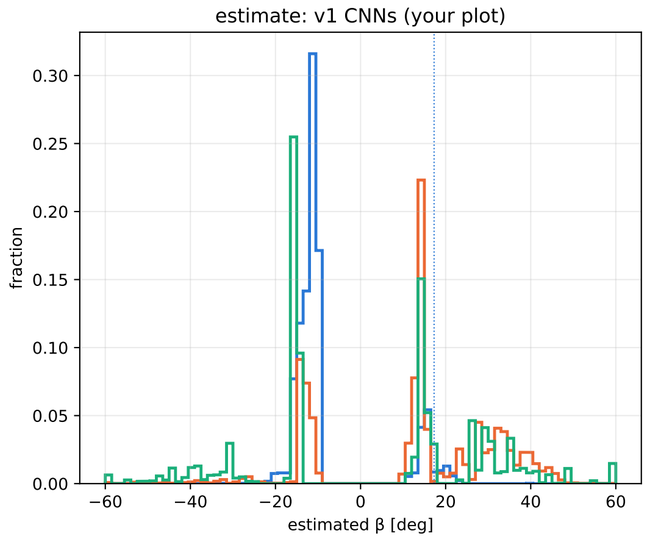










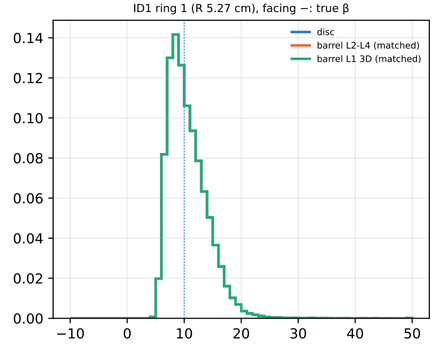


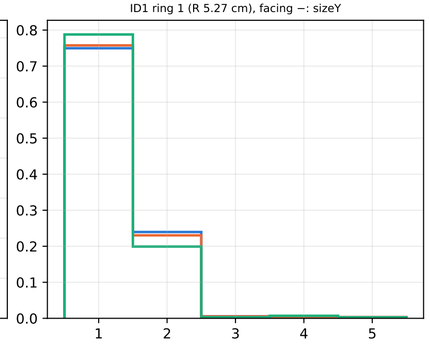
core RMS about 0 [deg] (full RMS for emuT / emuC)

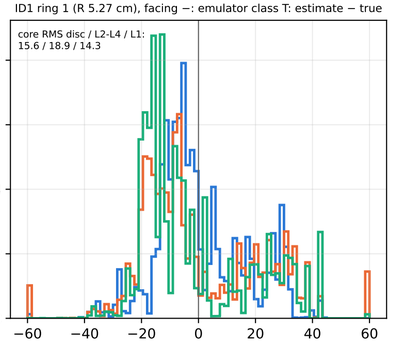
sample	Neff	P(sY≥2)	emuT	emuC	v2	v1sep	v1uni	full T	full C
disc	4569	0.397	18.0	19.0	2.2	27.8	25.8	18.9	20.1
L2-L4	1989	0.403	20.5	25.2	13.9	19.0	20.3	20.7	25.5
L1	210	0.365	20.2	35.2	23.3	30.0	33.4	20.5	35.3

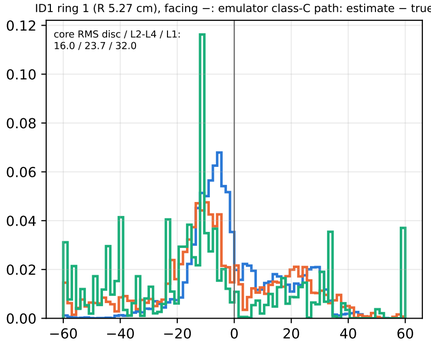
best raw L2-L4 sensor: L4 module 4 zOut +1
  (W1 9.6 deg, N 5081)
best raw L1 sensor: L1 module 4 zOut +1
  (W1 30.9 deg, N 13426)
disc (β, α) weight covered: L2-L4 1.000, L1 1.000
P(sY≥2): all 200 events;
Neff and resolutions: the 50 test events

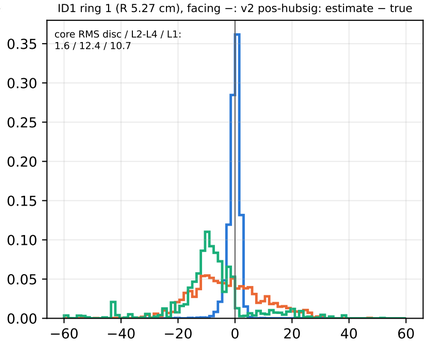
Pages 5-6. Two radii per disc (ring 1 = 5.3 cm, ring 4 = 13.8 cm), facing − (β > 0), vs barrel L2-L4 / L1 at matched incidence: residuals (estimate − true). pT ≥ 1 GeV. sizeY / true β: 200 events; estimates: 50 test events.

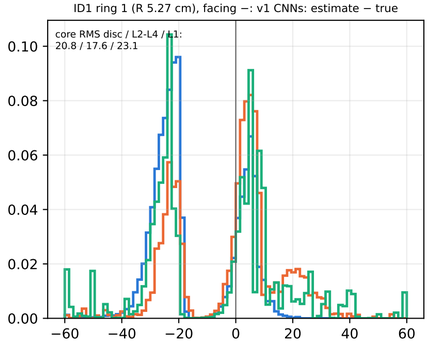


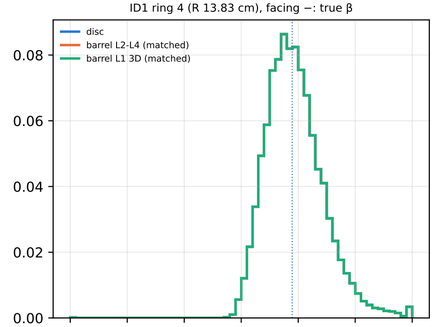


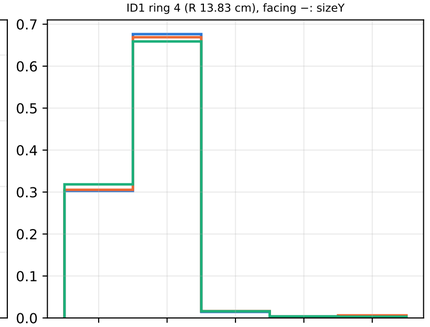


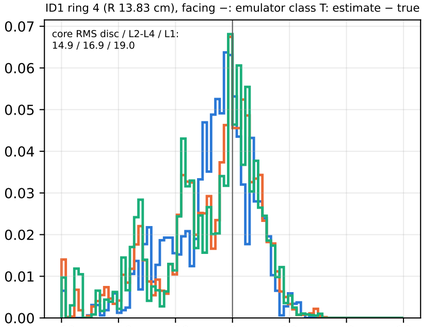


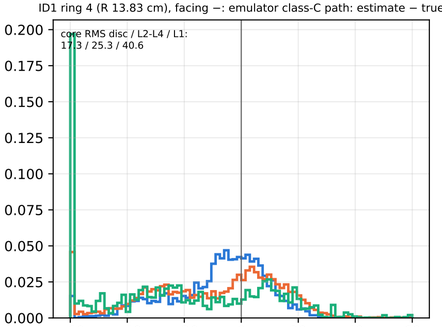


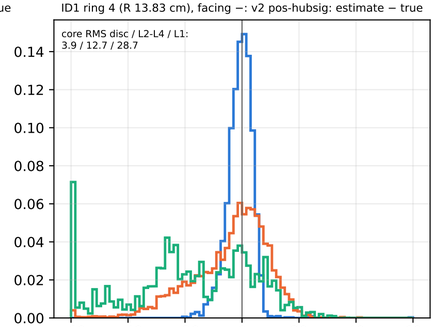


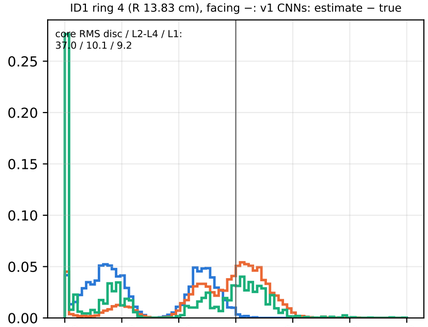


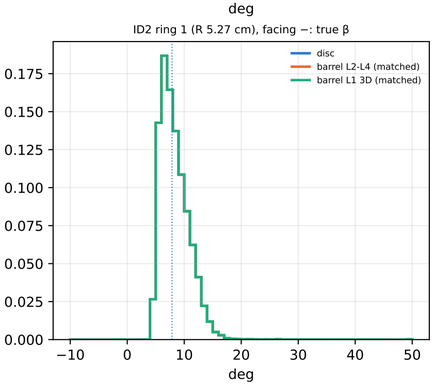


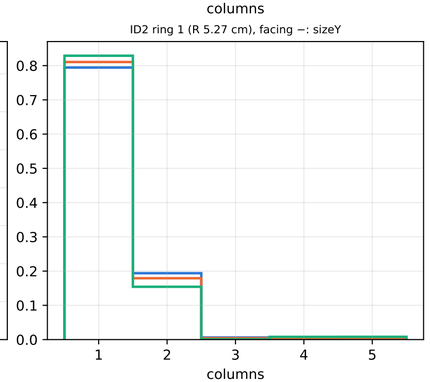


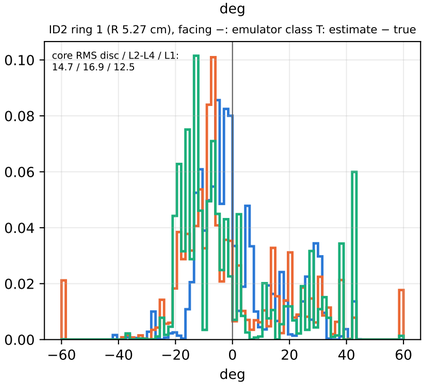


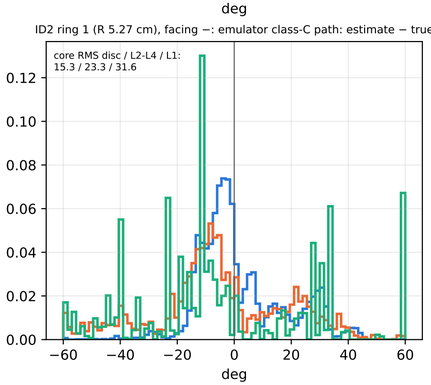


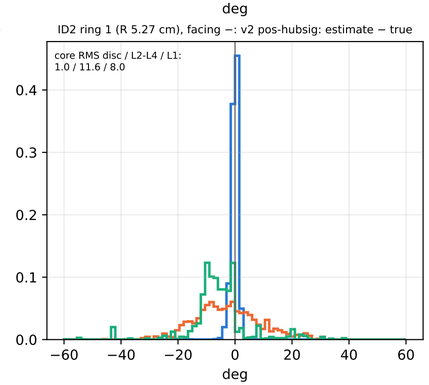


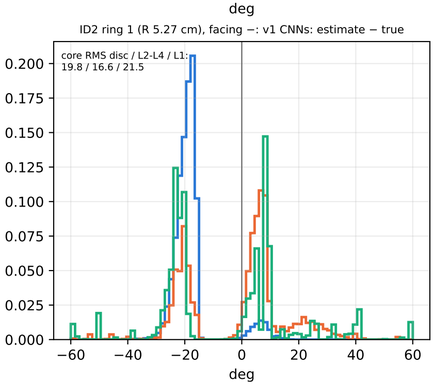


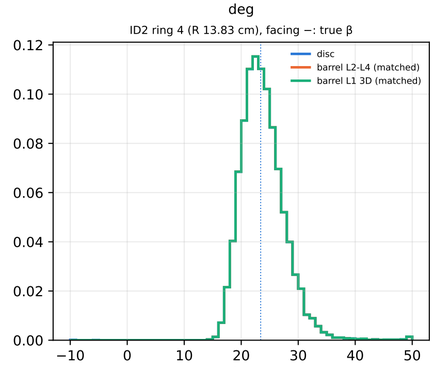


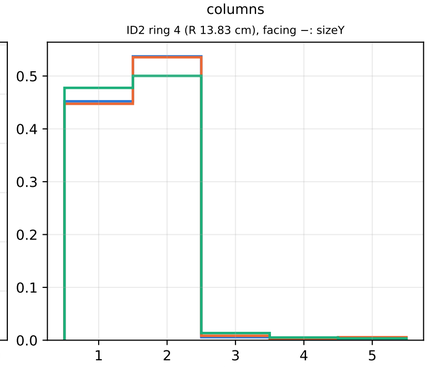


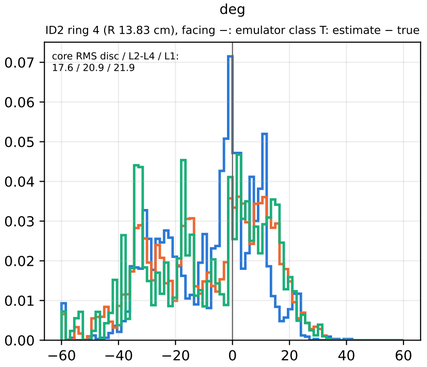


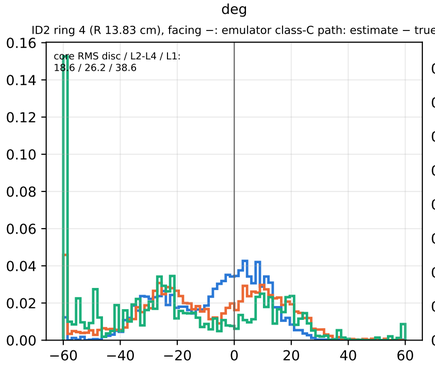


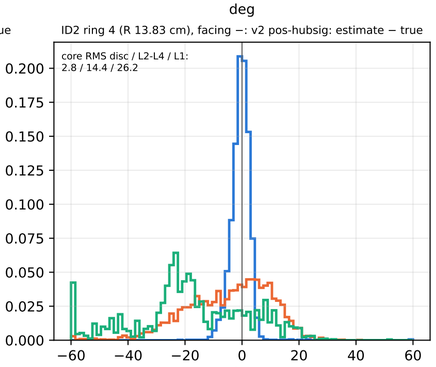


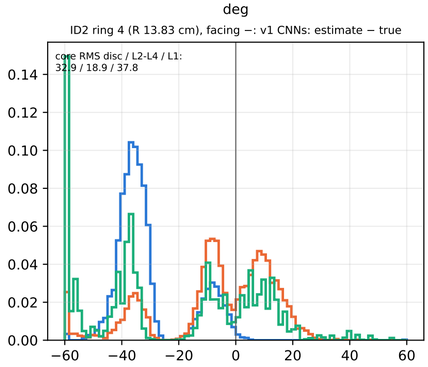




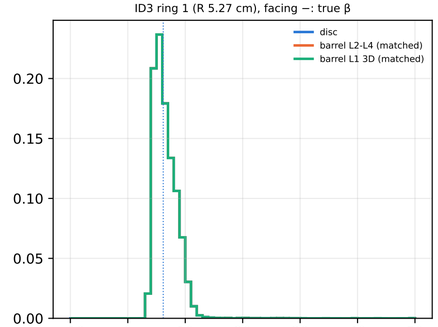


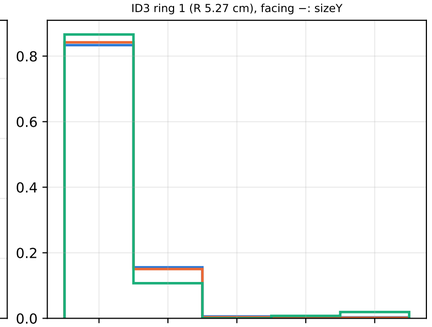


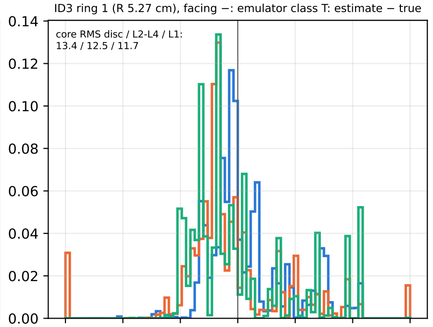


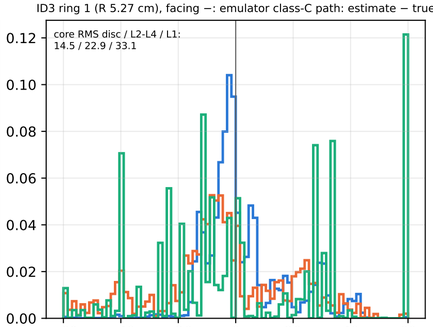


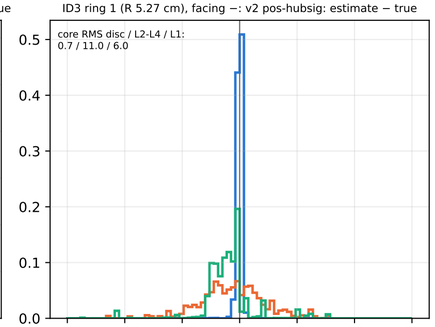
Pages 5-6. Two radii per disc (ring 1 = 5.3 cm, ring 4 = 13.8 cm), facing − (β > 0), vs barrel L2-L4 / L1 at matched incidence: residuals (estimate − true). pT ≥ 1 GeV. sizeY / true β: 200 events; estimates: 50 test events.

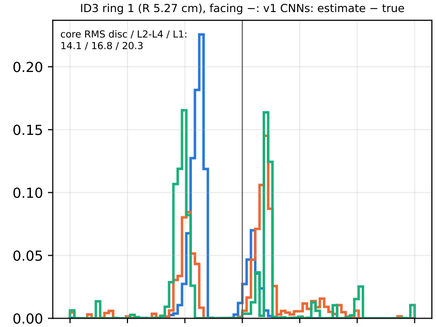


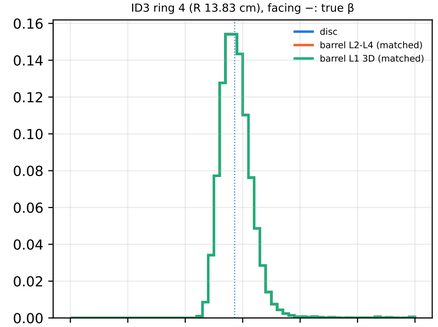


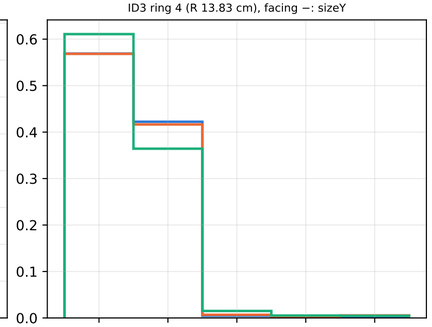


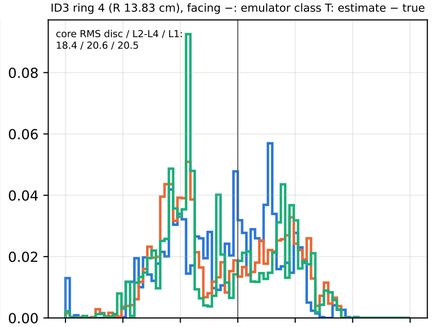


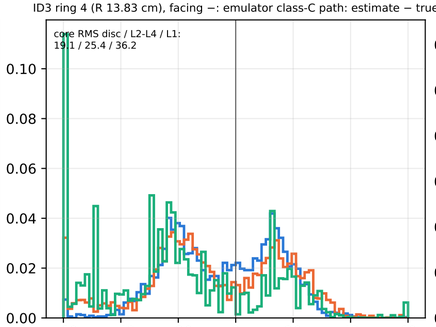


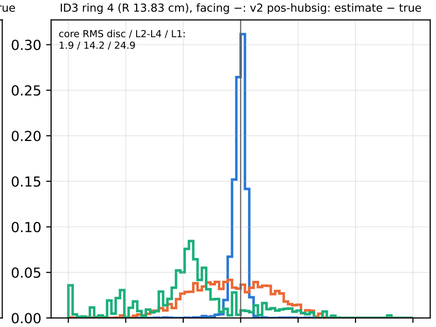


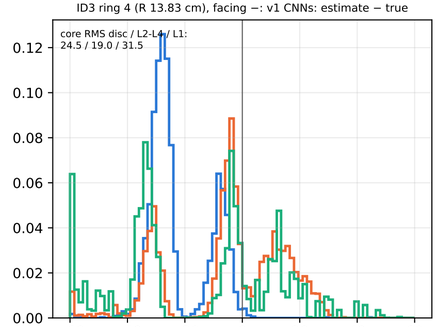


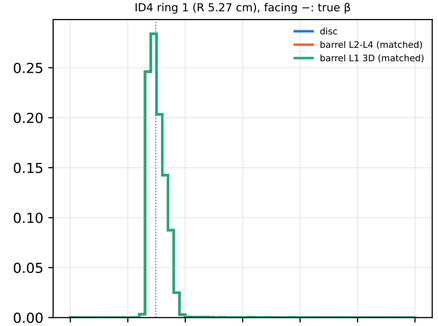


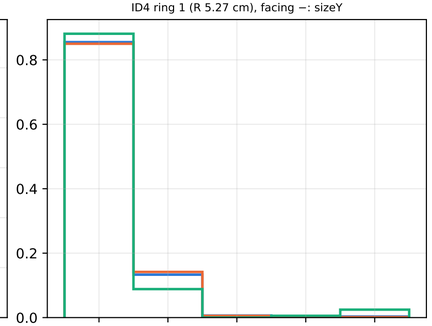


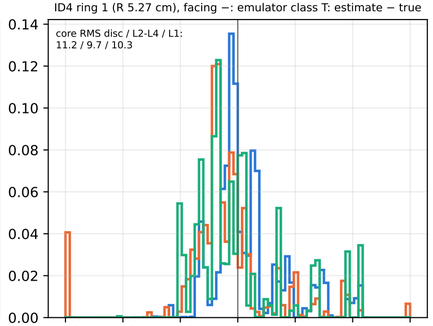


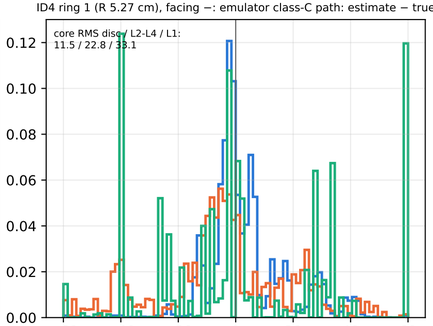


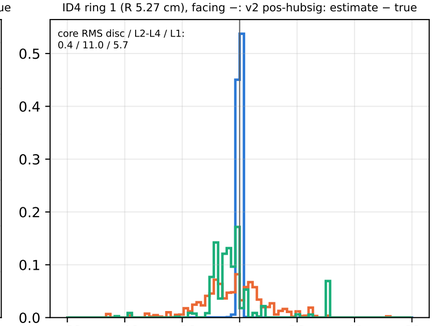


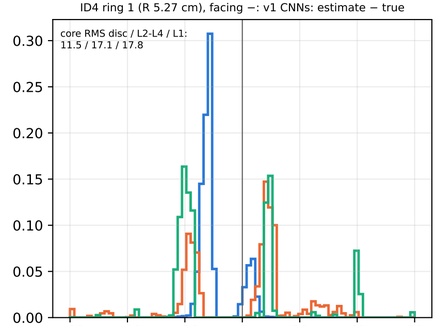


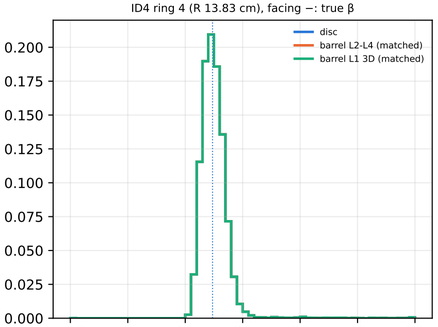


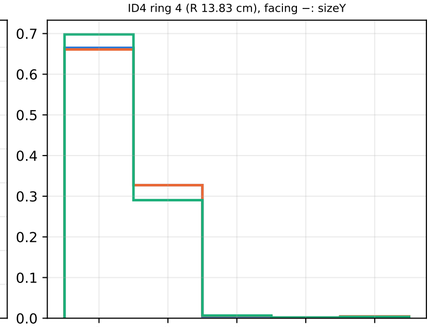


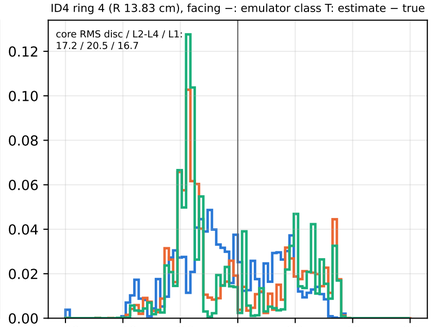


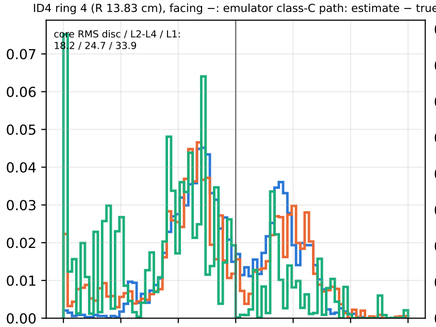


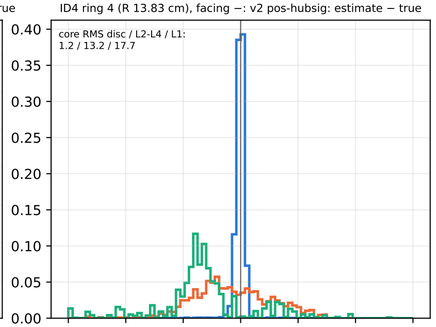


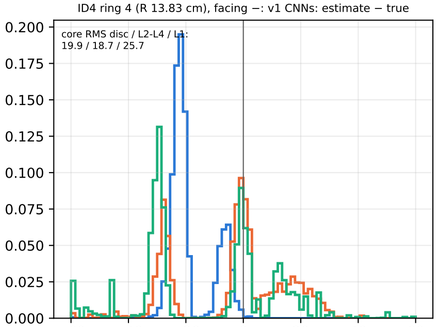




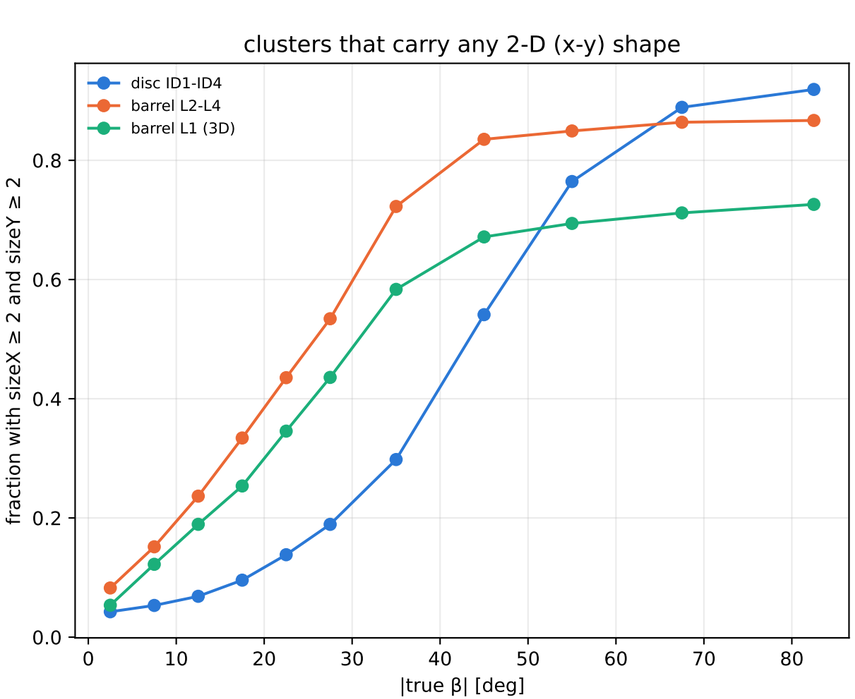


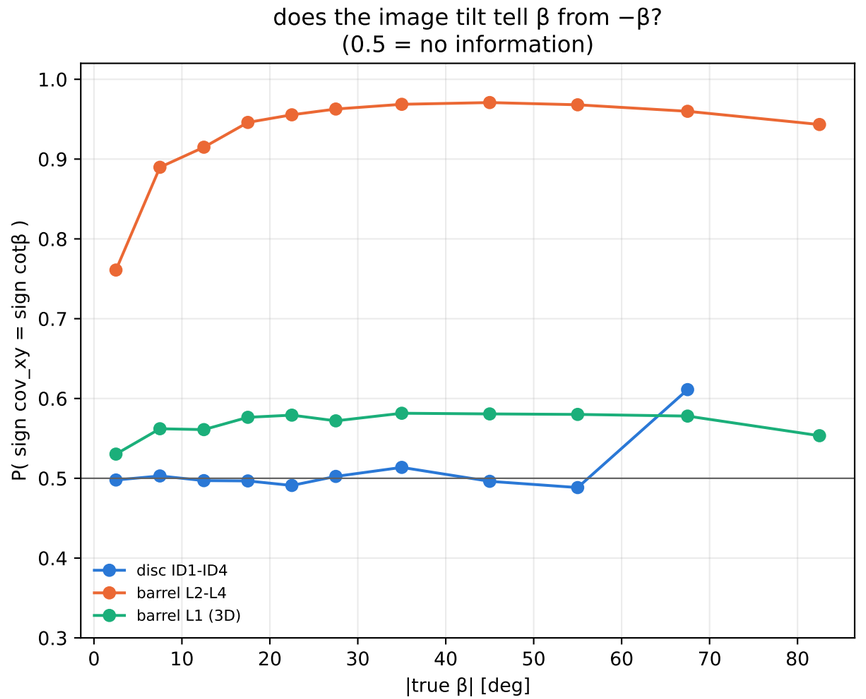


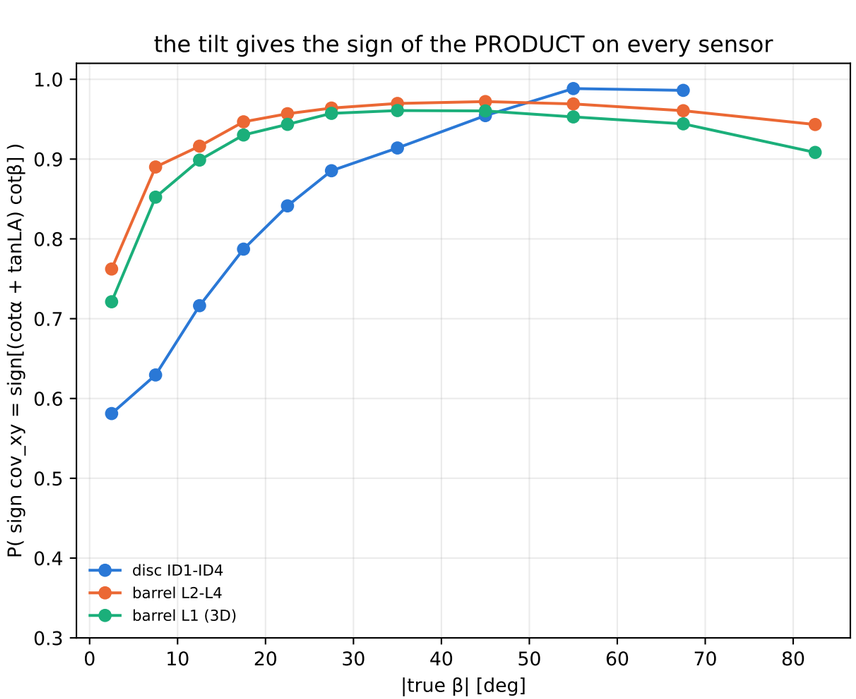




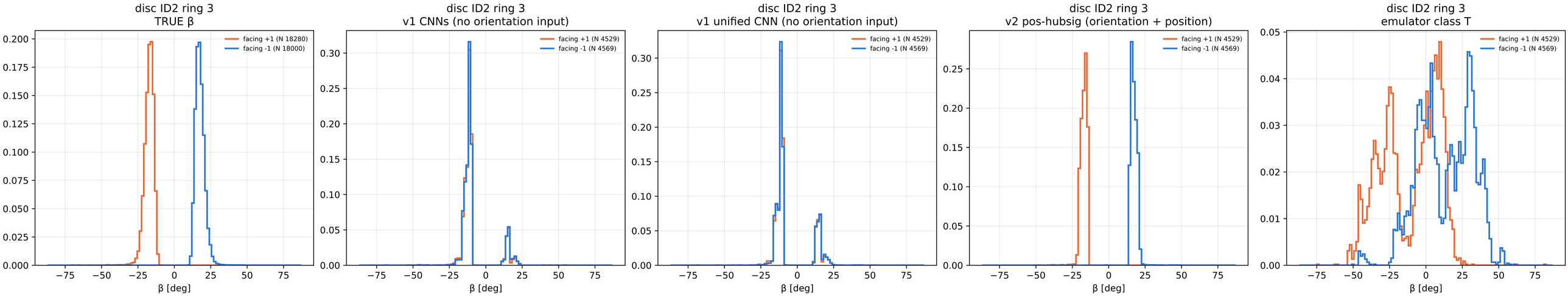
Appendix A. Why image-only nets resolve disc β worse near normal: a straight track deposits a segment along (cotα + tanLA, cotβ) in the (x, y) image. Without drift the image is the same for (α, β) and (-α, -β), so β and −β cannot be told apart and an image-only estimate of β ~ ±17° is off by ~2|β| half the time. With drift (L2-L4) the tilt separates them. Charge-weighted covariance of the 24x16 ADC image.

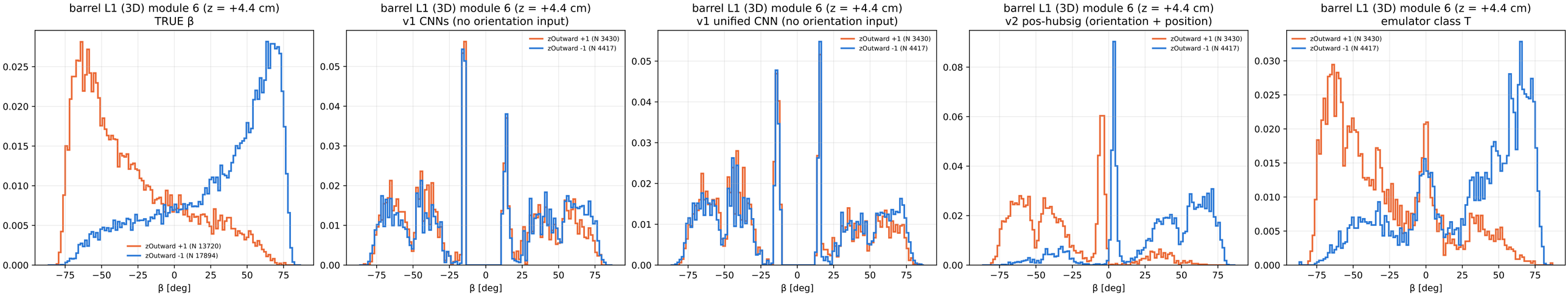






Appendix B. The two orientations of the same ring / module ring: the truth is mirrored (disc: β = −facing × R/|Z|). A net that cannot see the orientation, on a sensor whose image cannot tell β from −β (discs, L1), gives the same output for both, so its resolution is fine on one orientation and ~2|β| on the other. Estimates: 50 test events.





Appendix B (table). Resolution per surface × orientation (facing on the discs, zOutward on the barrel): core / full RMS of estimate − true [deg], test events

surface	orientation	N	med_true_beta	emuT_core	emuT_full	emuC_core	emuC_full	v2_core	v2_full	v1sep_core	v1sep_full	v1uni_core	v1uni_full
ID1	1	23237	-15.1	20.0	20.1	20.5	21.0	2.4	3.2	21.3	21.5	6.5	19.5
ID1	-1	23051	15.3	16.5	17.1	17.2	18.1	2.5	3.3	26.7	26.9	28.4	28.5
ID2	1	23541	-11.9	20.0	20.0	20.3	20.8	1.6	2.5	5.6	12.1	14.1	15.0
ID2	-1	23435	12.1	16.2	16.6	16.8	17.6	1.6	2.5	25.1	24.6	22.9	23.0
ID3	1	23282	-9.4	19.5	19.5	19.7	20.2	1.0	2.2	11.5	11.9	4.5	11.2
ID3	-1	23406	9.5	15.7	16.0	16.2	16.7	1.1	2.0	18.0	18.2	18.5	18.7
ID4	1	23155	-7.4	18.6	18.7	18.7	19.3	0.6	1.8	8.6	9.2	3.8	6.9
ID4	-1	22923	7.5	14.9	15.0	15.3	15.7	0.7	1.8	14.5	14.7	16.2	16.0
IL1	1	21112	1.5	3.4	12.2	8.8	26.9	3.9	18.9	70.7	70.7	70.9	70.9
IL1	-1	26700	-1.0	2.7	12.0	6.1	29.1	3.0	20.4	83.9	83.9	83.3	83.3
IL2	1	16269	-0.2	8.0	14.7	14.7	21.8	7.3	12.0	7.0	30.8	6.8	31.2
IL2	-1	18086	-0.7	7.6	14.2	13.1	21.3	6.4	12.1	6.3	32.1	6.5	32.1
IL3	1	12025	-0.4	12.3	16.1	17.7	21.7	8.6	10.5	11.3	28.4	12.2	28.3
IL3	-1	13130	-0.7	12.0	15.6	17.3	21.2	8.3	10.9	10.5	29.8	10.8	29.8
IL4	1	9775	-0.4	14.1	17.2	19.3	21.8	8.7	9.7	14.6	25.0	14.7	25.1
IL4	-1	10165	0.0	14.1	17.0	19.1	21.5	8.6	9.8	14.3	24.8	14.5	25.1

Appendix C. The SmartPixels dump (Conv1D_Full-2bit, CMS axes, 1x) at the two lookup keys. Class T emitted β = truth + this response with its spread x1.5 (bias kept). Discs are looked up at alpha_eq = atan(cotA - 0.312), where the PixelAV drift cancels (no tilt, as on a real disc); the barrel at the true alpha (~0 near normal, the drift tilts the cluster). Below |β| ~ 20 deg both keys sit on a plateau (estimate ~ a few deg whatever β is); at the disc key the plateau is centred a few deg POSITIVE, at the barrel key near 0, so the disc resolution is better for β > 0 and worse for β < 0 than the barrel's. Its sign in the CMS frame rests on the dump-to-CMS axis map (z_pix = -w_cms). Cells hold 40-115 dump rows.

	alpha_band	true_beta	N	median_estimate	median_residual	core_RMS	full_RMS
disc key	alpha_eq -21..-13 (no tilt)	-30..-20	110	-18.67	6.46	14.75	14.75
disc key	alpha_eq -21..-13 (no tilt)	-20..-12	74	-3.32	13.25	15.51	15.51
disc key	alpha_eq -21..-13 (no tilt)	-12..-6	56	-0.64	6.63	13.86	13.86
disc key	alpha_eq -21..-13 (no tilt)	-6..0	51	2.57	4.65	6.85	9.23
disc key	alpha_eq -21..-13 (no tilt)	0..6	57	4.46	2.55	9.11	9.73
disc key	alpha_eq -21..-13 (no tilt)	6..12	74	4.91	-3.54	9.35	9.35
disc key	alpha_eq -21..-13 (no tilt)	12..20	66	9.15	-5.64	12.24	13.23
disc key	alpha_eq -21..-13 (no tilt)	20..30	113	21.59	-3.61	12.14	12.14
barrel key	alpha -4..4 (drift tilt)	-30..-20	115	-23.01	2.64	13.46	14.84
barrel key	alpha -4..4 (drift tilt)	-20..-12	69	-7.26	8.21	14.13	14.13
barrel key	alpha -4..4 (drift tilt)	-12..-6	39	-4.24	3.63	11.13	11.13
barrel key	alpha -4..4 (drift tilt)	-6..0	38	0.08	2.43	4.15	6.84
barrel key	alpha -4..4 (drift tilt)	0..6	49	-0.25	-2.76	5.31	10.77
barrel key	alpha -4..4 (drift tilt)	6..12	61	0.92	-6.90	12.06	13.17
barrel key	alpha -4..4 (drift tilt)	12..20	72	2.29	-12.47	14.91	14.91
barrel key	alpha -4..4 (drift tilt)	20..30	104	20.16	-6.51	15.19	15.93

Appendix D. Per disc sensor (disk, ring, facing) vs matched barrel: resolution of estimate − true [deg] (core RMS about 0, fraction outside the core, full RMS); Neff of the test events

disk	ring	fac	sample	Neff	medβ	P(sY≥2)	emuT_core	emuT_out	emuT_full	emuC_core	emuC_full	v2_core	v1sep_core	v1sep_full
2	3	-1	disc	4569.000	17.283	0.397	18.011	0.009	18.902	19.032	20.091	2.169	27.788	27.897
2	3	-1	L2-L4	1988.571	17.286	0.403	20.492	0.001	20.705	25.173	25.524	13.888	18.991	19.968
2	3	-1	L1 3D	209.920	17.298	0.365	20.240	0.002	20.520	35.237	35.348	23.267	29.985	30.098
1	1	1	disc	8572.000	-9.777	0.246	20.109	0.000	20.109	20.585	20.995	1.622	15.859	16.005
1	1	1	L2-L4	1184.103	-9.807	0.242	17.263	0.000	17.310	23.780	24.157	12.779	20.931	21.461
1	1	1	L1 3D	119.072	-9.635	0.234	17.509	0.001	17.643	25.511	26.153	14.018	22.424	22.638
1	1	-1	disc	8570.000	9.997	0.250	15.561	0.001	15.715	16.015	16.359	1.636	20.801	20.914
1	1	-1	L2-L4	1225.092	9.996	0.242	18.930	0.025	21.273	23.699	24.104	12.450	17.629	17.928
1	1	-1	L1 3D	123.658	9.973	0.212	14.317	0.237	19.358	31.964	31.964	10.700	23.081	24.897
1	4	1	disc	3290.000	-28.428	0.681	17.518	0.001	17.825	18.274	18.551	3.727	31.369	31.406
1	4	1	L2-L4	5071.426	-28.423	0.677	16.172	0.012	18.117	22.643	24.157	12.666	9.699	25.145
1	4	1	L1 3D	650.712	-28.399	0.662	15.307	0.016	17.789	38.452	38.541	29.468	43.638	43.640
1	4	-1	disc	3217.000	28.946	0.697	14.942	0.013	16.594	17.302	19.774	3.899	36.996	37.558
1	4	-1	L2-L4	5155.016	28.935	0.695	16.858	0.023	19.590	25.324	26.395	12.742	10.059	22.794
1	4	-1	L1 3D	606.049	28.930	0.682	18.964	0.010	20.303	40.580	40.589	28.724	9.183	44.052
2	1	1	disc	8532.000	-7.720	0.204	18.933	0.000	18.933	19.299	19.463	1.009	4.261	7.687
2	1	1	L2-L4	698.835	-7.720	0.188	15.874	0.000	15.878	23.149	24.016	12.171	19.243	20.036
2	1	1	L1 3D	59.111	-7.667	0.176	10.325	0.147	15.502	22.734	23.225	9.397	18.225	19.333
2	1	-1	disc	8572.000	7.858	0.206	14.740	0.000	14.774	15.290	15.500	1.047	19.831	19.549
2	1	-1	L2-L4	709.587	7.889	0.189	16.911	0.039	20.777	23.276	23.894	11.614	16.553	17.109
2	1	-1	L1 3D	59.157	7.871	0.171	12.533	0.231	19.669	31.609	31.617	8.040	21.473	23.728
2	4	1	disc	3365.000	-22.999	0.536	20.192	0.000	20.260	20.519	20.831	2.768	20.313	20.724
2	4	1	L2-L4	3012.954	-22.997	0.538	20.246	0.009	21.391	24.173	24.917	13.957	11.468	25.215
2	4	1	L1 3D	320.290	-22.993	0.522	19.064	0.022	21.814	37.130	37.153	27.200	37.946	37.990
2	4	-1	disc	3329.000	23.411	0.548	17.550	0.010	18.671	18.605	20.024	2.813	32.929	33.149
2	4	-1	L2-L4	3237.800	23.407	0.553	20.871	0.005	21.578	26.168	26.754	14.402	18.919	21.826
2	4	-1	L1 3D	349.363	23.360	0.522	21.856	0.006	22.681	38.570	38.573	26.227	37.765	37.795
3	1	1	disc	8345.000	-6.077	0.170	12.436	0.175	17.685	12.486	18.013	0.633	8.424	8.840
3	1	1	L2-L4	440.621	-6.106	0.142	7.109	0.196	14.903	22.301	23.316	11.359	17.707	18.364
3	1	1	L1 3D	30.680	-6.024	0.132	7.999	0.102	12.728	21.924	22.370	7.597	8.818	17.429
3	1	-1	disc	8371.000	6.173	0.166	13.369	0.017	14.380	14.473	14.907	0.658	14.135	14.315
3	1	-1	L2-L4	409.810	6.236	0.158	12.481	0.079	19.353	22.898	23.701	10.960	16.785	17.680
3	1	-1	L1 3D	40.191	6.216	0.134	11.675	0.156	17.863	33.144	33.152	5.991	20.319	22.268
3	4	1	disc	3402.000	-18.374	0.423	20.095	0.000	20.095	20.847	21.145	1.822	17.623	17.890
3	4	1	L2-L4	1603.859	-18.370	0.427	20.857	0.005	21.509	24.657	24.972	14.265	24.304	24.530
3	4	1	L1 3D	171.975	-18.340	0.410	21.204	0.017	23.306	33.261	33.292	20.701	35.211	35.245
3	4	-1	disc	3389.000	18.605	0.431	18.399	0.013	19.616	19.068	19.894	1.890	24.545	24.735
3	4	-1	L2-L4	1710.730	18.649	0.432	20.565	0.002	20.796	25.438	25.790	14.205	18.969	20.268
3	4	-1	L1 3D	175.177	18.631	0.389	20.480	0.002	20.825	36.215	36.396	24.897	31.541	31.545
4	1	1	disc	8086.000	-4.795	0.143	10.021	0.155	15.608	9.950	16.075	0.395	6.451	6.876
4	1	1	L2-L4	307.595	-4.800	0.113	6.125	0.145	13.434	22.250	23.079	10.814	16.607	16.914
4	1	1	L1 3D	20.319	-4.818	0.114	5.891	0.104	10.638	18.491	23.171	6.360	15.798	15.944
4	1	-1	disc	8144.000	4.867	0.145	11.157	0.042	13.719	11.515	14.386	0.415	11.473	11.724
4	1	-1	L2-L4	279.731	4.887	0.150	9.653	0.106	18.120	22.828	23.340	11.009	17.093	18.232
4	1	-1	L1 3D	28.454	4.737	0.119	10.318	0.123	15.684	33.113	33.114	5.666	17.814	21.698
4	4	1	disc	3478.000	-14.588	0.325	21.141	0.000	21.141	21.283	22.464	1.183	13.358	13.798
4	4	1	L2-L4	927.899	-14.561	0.325	20.043	0.000	20.088	24.473	24.580	13.667	22.813	23.211
4	4	1	L1 3D	104.880	-14.566	0.351	20.310	0.002	20.583	29.896	29.947	17.106	27.348	27.461
4	4	-1	disc	3411.000	14.756	0.335	17.228	0.004	17.658	18.162	18.564	1.185	19.948	20.120
4	4	-1	L2-L4	909.854	14.746	0.339	20.504	0.000	20.518	24.663	24.823	13.159	18.682	19.138
4	4	-1	L1 3D	116.960	14.855	0.302	16.745	0.312	20.067	33.863	33.907	17.655	25.739	26.603